



Reg. No:

--	--	--	--	--	--	--	--	--	--	--

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

COMMON TO B.E – CSE / B.TECH – IT / AIDS & M.TECH CSE

20IT204 - COMPUTER ORGANIZATION AND ARCHITECTURE

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1** Which component is responsible for fetching instructions in a computer system?
a) ALU b) Control Unit c) Register d) Cache
2. **CO2** In a 32-bit computer, how many bits are used to represent an address?
a) 4 b) 8 c) 16 d) 32
3. **CO4** What does program counter (PC) store?
a) Address of the current instruction b) Address of the next instruction
c) Program data d) Address of the I/O devices
4. **CO2** The ALU performs a bitwise OR operation on two binary numbers, 1010 and 1100. What is the result?
a) 1010 b) 1100 c) 1110 d) 1000
5. **CO3** Which memory has the fastest access time?
a) Cache b) RAM c) Hard disk d) ROM
6. **CO5** What does the term 'pipeline' in computer architecture refer to?
a) Data Storage b) Memory allocation
c) Instruction overlap d) Bus communication

7. **CO6** In dynamic scheduling, which unit is responsible for deciding the order of instruction execution based on available resources?
a) ALU b) Control Unit c) Register file d) Dispatch Unit

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO2** Which register in the CPU indicates if the last operation was zero, negative, or had a carry?
9. **CO3** What component in a CPU temporarily holds the data being transferred to or from a peripheral?
10. **CO6** ILP refers to _____.

PART - B

(5x 3 = 15 Marks)

11. **CO1** Differentiate between RISC and CISC architectures.
12. **CO1** Explain any three addressing modes.
13. **CO3** Present the significance of virtual memory.
14. **CO4** Demonstrate hardwired control unit.
15. **CO6** Explain Multicore.

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO2** i) Design a fast circuit for adding two 8-bit numbers resulting in minimal gate delays. (10)
CO2 ii) Apply Booth's algorithm in multiplying 22×-10 (5)
17. **CO2** i) Perform non-restoring division $20 / 8$ and find how quotient and remainder are obtained (8)
CO2 ii) Find the floating-point representation of (-75.25) in IEEE 754 32 bit format (7)

18. **CO5** Consider the following sequence of instructions in a simplified pipeline (15)
that operates with Tomasulo's algorithm for dynamic scheduling:

Instruction	Operation	Source 1	Source 2	Destination
I1	ADD	R1	R2	R3
I2	MUL	R3	R4	R5
I3	SUB	R5	R6	R7
I4	DIV	R7	R8	R9

Assume that:

- The machine has three functional units: an adder (2 cycles), a multiplier (4 cycles), and a divider (8 cycles).
- There are 2 reservation stations available for each type of operation.
- The latency for register read/write operations is 1 cycle, and instruction issue takes 1 cycle.
- Register values are initially valid except where dependencies arise between instructions.

Show the Tomasulo's scheduling with an explanation of dealing with register renaming and dependencies.

19. **CO4** i) A centralized bus master is recommended in a I/O device connections (10)
system that wants a higher response time though the cost may be high.
Analyze different methods of implementing a centralized bus arbiter and
suggest a suitable one for the given requirement
- CO4** ii) Present the distributed arbitration with an example (5)
20. **CO3** i) Assume a main memory of size 4 GB and a cache size of 4 MB. Both the (10)
memory is divided with 8 bytes blocks. Demonstrate the design of 4-way
set associative cache mapping with a neat diagram.
- CO3** ii) Illustrate the different write principles of cache if a single cache is used. (5)
21. **CO5** i) Discuss on pipelining implementation (7)
- CO5** ii) Explain any two pipeline hazards (8)

22. **CO4** Sub R1, ((R3)); // Subtract the memory content pointed to by address in (15)
the address contained in R3 (i.e. R3 – indirect mode) from Register R1
Generate micro-operations for the fetching of the given instruction and
execution by providing suitable explanation
